

# Reviewer Pack — Minimal Local Indefinite Integral and Communication Complexi...

Entry 7562120dfe88 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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## Minimal Local Indefinite Integral and Communication Complexity Rank via Geometric Langlands

Entry ID: 7562120dfe88 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 15:04:00 UTC

### 1. Conjecture

**Field A** (mathematical branch): Geometric Langlands Correspondence Theory **Field B** (complexity object): Communication Complexity

#### Statement:

For any given communication protocol  $\varphi$ , the minimal local indefinite integral (LII) of its associated vector bundle in the geometric Langlands correspondence is linearly correlated with its communication complexity rank  $r(\varphi)$ , such that  $LII(\varphi) = \Theta(r(\varphi))$ . Furthermore, for all instances with property P, if  $r(\varphi) \leq C * |S|$ , then the  $LII(\varphi) \leq D * |S|$ , where  $|S|$  is the size of the instance and  $C, D$  are constants.

#### Rationale (proposer's reasoning):

The geometric Langlands correspondence provides a bridge between algebraic geometry and number theory. The minimal local indefinite integral captures the geometric structure of vector bundles, which may encode information about computational complexity. By linking this to communication complexity rank, we might uncover new insights into the nature of computation.

**Taxonomy category:** Geometric\_Langlands (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix):** cf800713b54065c1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

#### Acceptance criterion:

The conjecture is supported if the correlation coefficient between  $LII(\varphi)$  and  $r(\varphi)$  is  $\geq 0.8$  for at least 80% of the seeds, with all seed-generated LII values  $\leq 2 * |S|$  AND all seed-generated  $r(\varphi)$  values  $\leq 5 * |S|$ .

### 3. Barrier filter (F1)

**Final:** PASS

| Barrier        | Verdict   | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|----------------|-----------|------------|------------------|------------------|
| RELATIVIZATION | UNCERTAIN | 0.90       | UNCERTAIN        | HITS             |
| NATURAL_PROOFS | SAFE      | 0.50       | UNCERTAIN        | UNCERTAIN        |
| ALGEBRIZATION  | SAFE      | 0.50       | UNCERTAIN        | UNCERTAIN        |
| KARP_LIPTON    | SAFE      | 0.50       | UNCERTAIN        | UNCERTAIN        |

### 4. Novelty audit

**Verdict:** NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

### 5. Empirical test harness

**Multi-seed protocol:** 5 seeds (11, 23, 37, 53, 71)

**Sandbox:** isolated subprocess, stdlib-only,  $\leq 90$ s wall time per cycle **Execution:** rc=1, elapsed=0.1s

#### 5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Simulate generating a random communication protocol  $\phi$  with size  $n \leq 40$ 
    n = random.randint(5, 40)
    communication_protocol = [random.random() for _ in range(n)]

    # Simulate computing the associated vector bundle (this is a placeholder)
    vector_bundle = [[random.random() for _ in range(n)] for _ in range(n)]

    # Simulate determining the minimal local indefinite integral  $LII(\phi)$  using geometric Langlands
    # This is a placeholder function that returns a random value
    def compute_LII(vector_bundle):
        return sum(sum(row) for row in vector_bundle)

    LII = compute_LII(vector_bundle)

    # Simulate measuring the communication complexity rank  $r(\phi)$ 
    # This is a placeholder function that returns a random value
    def compute_communication_complexity_rank(communication_protocol):
        return len([x for x in communication_protocol if x > 0.5])

    communication_complexity_rank = compute_communication_complexity_rank(communication_protocol)

    # Check the acceptance criterion
    C, D = 1, 2 # Example constants
    size_of_instance = n

    conjecture_holds = False
    counterexample = ""
```

```

if LII <= 2 * size_of_instance and communication_complexity_rank <= 5 * size_of_instance:
    correlation_coefficient = random.random() # Placeholder for actual calculation
    if correlation_coefficient >= 0.8:
        conjecture_holds = True

return {
    "metric_name": "correlation_coefficient",
    "metric_value": correlation_coefficient,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, **{trial_result}}}")
        results.append(trial_result)

    mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
    std_metric_value = math.sqrt(sum((result["metric_value"] - mean_metric_value) ** 2 for result
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        counterexample = "mapping_undefined"
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

## 6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

## 7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60fb8262.py", line 69, in <module>
    trial_result = run_trial(seed)
    ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60fb8262.py", line 56, in run_trial
    "metric_value": correlation_coefficient,
    ~~~~~
UnboundLocalError: cannot access local variable 'correlation_coefficient' where it is not associated

```

## 8. Critic adversarial review

**Critic verdict:** CHALLENGE

**Critic reasoning:**

skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict:** INCONCLUSIVE

**Reasoning:**

The test crashed before producing data, which means it did not meet the pre-registered support condition and therefore cannot be supported or falsified. | next: Re-run the test to ensure it completes without crashing and produces the required data for analysis.

## 11. Audit log (LLM calls)

**Total LLM calls:** 10

| #  | Phase           | Provider      | Model            | Tokens in | out | Latency (ms) |
|----|-----------------|---------------|------------------|-----------|-----|--------------|
| 1  | propose         | ollama_remote | glm4:latest      | 0         | 0   | 15323        |
| 2  | propose         | ollama_remote | glm4:latest      | 0         | 0   | 43501        |
| 3  | preregistration | ollama_remote | glm4:latest      | 0         | 0   | 9794         |
| 4  | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 15927        |
| 5  | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 24840        |
| 6  | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 23688        |
| 7  | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 13174        |
| 8  | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 10507        |
| 9  | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 13777        |
| 10 | judge           | ollama_remote | glm4:latest      | 0         | 0   | 12173        |

**Totals:** 0 input tokens, 0 output tokens, ~\$0.0000 cost, 182706 ms total latency. Provider mix: {'ollama\_remote': 10}

*(full prompt+response transcripts available in research/audit/7562120dfe88.jsonl)*

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/7562120dfe88.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file:** research/pvsnp\_sandbox/test\_\*.py (per-cycle)

**Replay tarball:** research/replay/7562120dfe88.tar.gz (if generated)